CHAPTER I
Problems in the Classical Model of AI

Symbolic AI

Symbolic Artificial Intelligence, commonly known as classical AI or Good Old-Fashioned AI (GOFAI), constitutes one of the first and most significant methodologies for constructing intelligent machines. Originating in the mid-twentieth century, symbolic AI is predicated on the notion that intelligence can be attained through the manipulation of symbols in accordance with formal regulations. Within this perspective, cognition is regarded as a computational process, akin to executing logical operations on representations of reality. This method has dominated AI research for several decades and influenced philosophical discussions over the essence of mind and intelligence. It is sometimes referred to as classical artificial intelligence (AI) and GOFAI, an acronym for "Good Old-Fashioned AI", a term introduced by John Haugeland in 1985. This notion posits that brain activity resembles the sequential information processing executed by digital computers. Thinking is regarded as the manipulation of symbols and the computation of formal representations upon formal representations.

Symbolically, artificial intelligence is predicated on the core notion that intelligent behaviour directly arises from the ability to explicitly express information. Symbols represent objects, attributes, categories, and relations. These symbolic constructs can be amalgamated into propositions or assertions, which the system can subsequently analyse and modify through logical operations. An illustration of this would be software that contains the information "All humans are mortal" and "Socrates is a human," enabling it to deduce the conclusion "Socrates is mortal." This argument illustrates how the manipulation of formal symbols can provide outcomes that appear intelligent.

This strategy was inspired by findings in mathematical logic, linguistics, and early computer science. Allen Newell and Herbert Simon proposed the concept that the human mind operates similarly to a symbol-processing computer. Their renowned Physical Symbol System Hypothesis posits that any system capable of handling symbols appropriately may theoretically exhibit global intelligence. This notion not only endorsed the belief in the potential creation of machine minds but also provided a theoretical basis for ambitious artificial intelligence initiatives.

Systematic methods of knowledge representation are crucial to symbolic artificial intelligence. Scholars over their scientific careers have developed tools such as production rules, semantic networks, frames, and ontologies. Production rules sometimes manifest as an "if–then" structure, allowing systems to respond to certain conditions by executing the necessary tasks. Frames provide comprehensive elucidations of conventional situations, while semantic networks structure information based on the relationships among various concepts, such as categorising animals by their characteristics or linking events in a timeline. The application of these strategies allows machines to retain extensive information in an orderly and intelligible manner.

The development of expert systems stands as one of the most notable achievements of symbolic artificial intelligence. The decision-making abilities of human experts in fields like medicine, geology, and engineering inspired the creation of these various systems. Such systems could deliver diagnoses, recommendations, or forecasts by embedding expert information into rule-based frameworks. During the 1970s and 1980s, expert systems effectively showcased the practical applications of symbolic artificial intelligence, leading to significant economic interest, particularly in fields such as medicine, finance, and manufacturing where decision-making processes could be enhanced by automated reasoning.

Symbolic artificial intelligence has advanced in disciplines such as automated theorem proving, game playing, and planning. The development of these algorithms could yield mathematical proofs, enable competitive chess play, and devise strategies for attaining intricate objectives. The belief that intellect could be simplified to computation and adherence to rules was reinforced by these notable accomplishments.

Symbolic artificial intelligence is closely associated with functionalism, the philosophical perspective asserting that mental states are defined by the causal roles they fulfil rather than by the physical makeup of their constituents. If the organisation of actions is paramount, rather than the substance from which they originate, then it is conceivable that computers may execute the same functions as brains. Symbolic artificial intelligence (AI) established a basis by offering a paradigm for contemplating how artificial systems may embody ideas, aims, or intentions, which suggests that these systems could potentially replicate cognitive processes similar to those of human brains.

The concept of symbolic artificial intelligence posits that intelligence can be attained by manipulating symbols within a computer according to established protocols. The essential methodological premises that support the techniques of symbolic artificial intelligence

In symbolic approaches, there are two primary subcategories: artificial intelligence (AI). In the first category, researchers endeavour to develop universal models of intelligent behaviour and articulate knowledge in a generalised format. This collection encompasses both logic-based reasoning and cognitive simulation.

cognitive simulation

Cognitive simulation was among the initial methodologies employed in the domain of artificial intelligence. Newell and Simon introduced the Logic Theorist and General Problem Solver (GPS) as systems developed through this method to create their renowned frameworks. A fundamental concept of cognitive simulation is the creation of heuristic algorithms that can replicate human cognitive functions, including learning, problem-solving, reasoning, and object recognition. In such a simulation, a computer performs a sequence of essential tasks comparable to those executed by a human. Consequently, we seek the fundamental notions and principles that humans use to address universal problems, aiming to design corresponding algorithms.

Four fundamental ideas that underpin cognitive simulation are as follows:

state space,

problem-solving as searching state space

Means-Ends Analysis,

problem reduction

space of the state

The concept of a state space is fundamental in artificial intelligence (AI), especially for problem solving, searching, planning, and decision making. The term "state space" denotes the whole set of all possible configurations or conditions that an agent or system may occupy, along with the permissible transitions between these configurations and conditions. By articulating issues in terms of states and actions, artificial intelligence can systematically explore the potential pathways from an initial point to a desired endpoint.

The state denotes a characterisation of the present condition of the environment. The entirety of all potential states is referred to as the state space. Artificial intelligence systems must transition between states by employing operators or actions to function. The goal is often to delineate a sequence of actions that can transition from the present condition to the desired state.

Consider a game of chess at this juncture. Each conceivable arrangement of the pieces on the board represents a distinct state. All potential board configurations permitted under chess rules are encompassed inside the state space. Artificial intelligence algorithms must use efficient

search strategies and heuristics to traverse this astronomically large number. The execution of such a systematic study would be unfeasible without the existence of the concept of a state space. The problem-solving process commences from the initial state, which denotes the condition from which we initiate. Consider the game of chess as an illustration. The initial state of a game denotes the primary configuration of the pieces in play. The condition of a problem when a solution is identified is termed the 'goal state', or the 'singular goal state' when the solution is unique. The phrase "objective state" denotes any condition in which we can attain checkmate against our adversary in chess. The remaining intermediate states encompass all potential contingencies that may emerge during the problem-solving process, including various moves and strategies that players might consider before reaching the objective state of checkmate. Consequently, in chess, all valid possibilities are displayed in accordance with the game's rules. A state space is a graph where nodes signify states (starting, goal, and intermediate) and edges denote permissible transitions between these states. This illustrates how a state space for chess might be delineated from this viewpoint. Commencing with the node denoting the initial state, we delineate transitions (edges) to nodes that reflect the outcomes following the first "white" step. Subsequently, we delineate transitions (edges) to nodes representing scenarios that arise following the initial "black" move in reaction to a condition instigated by the first "white" move, and so on. This is executed for every intermediate phase.

The traditional formulation of an issue that utilises state space comprises several distinct components. The initial state is a depiction of the agent's beginning place. Success is characterised by the intended outcome, which is the second element. Subsequently, we have the actions or operators that facilitate the transformation of one state into another, which can include strategies, decisions, or processes that lead to achieving the intended outcome. The path cost is the fourth criterion to evaluate, as it aids in identifying the most optimal alternative among multiple available paths. To effectively investigate this structure, artificial intelligence algorithms such as Breadth-First Search, Depth-First Search, A*, and others have been devised. The state space technique is advantageous beyond mere games. A state in robotics encompasses the robot's position, orientation, and sensor readings. A state may represent a city or location in route planning considerations, such as the starting point or destination in a navigation system. The 8-puzzle and Rubik's Cube are two instances of puzzle problems when each potential configuration of tiles or colours represents a state. The artificial intelligence will explore the state space until it identifies the suitable configuration.

The 8-puzzle exemplifies state space search and problem-solving methodologies. The board is three by three inches and contains eight numbered tiles along with one blank space. The goal of the game is to manoeuvre the tiles by sliding them into the vacant space until the board aligns with a designated target configuration, which typically consists of numbers arranged in a precise sequence.

Each conceivable arrangement of the tiles in the 8-puzzle represents a distinct state. The complete set of all viable configurations constitutes the state space. An initial state functions as the commencement point for the problem, with the aim of navigating through a sequence of permissible actions to reach the destination state.

A tile adjacent to the blank spot can slide into it. The operators in this problem are uncomplicated. There may be two, three, or even four potential moves, contingent upon the position of the blank. Each operation produces a new state that is incorporated into the search tree.

Utilising search algorithms is a standard approach to addressing challenges associated with the 8-Puzzle. Uninformed approaches such as Breadth-First Search or Depth-First Search can be utilised; nevertheless, these methods may explore numerous extraneous states. Consequently, suggested search tactics encompass A* and various other informed search methodologies. Heuristic functions are employed to estimate the proximity of a state to the aim.

Moreover, the notion of state space holds comparable importance in the domain of reinforcement learning. A reward is bestowed onto the agent upon its observation of the current state of the environment, subsequent decision-making regarding an action, and subsequent transition to a new state. The agent learns by identifying which actions under specific conditions yield superior outcomes, such as maximising rewards or minimising penalties in various scenarios. The capacity to effectively traverse the state space is, thus, the foundation of intelligence.

Conversely, combinatorial explosion presents a substantial challenge. In numerous real-world challenges, the number of states might increase significantly. Consequently, it is an unfeasible endeavour to examine every possibility. Artificial intelligence employs heuristics, abstraction, approximation, and probabilistic reasoning to minimise search efforts.

The notion of state space links artificial intelligence with cognitive science and philosophy. Some scholars argue that we can understand human cognition as navigating mental state spaces, evaluating alternatives, and selecting optimal actions. Thus, AI emulates human problem-solving capabilities.

problem-solving as searching state space

This represents the second concept of cognitive simulation, pertaining to problem-solving as a search inside a state space. The concept is simple: if we cannot elucidate algorithmic principles for problem-solving, we might address the issue by the method of "trial and error" (sometimes referred to as "make and test" or "guess and check"). Nonetheless, there are instances when this is the sole alternative available for us to engage in our daily tasks, such as when we are faced with challenges like remembering the combination code for our bags, which we have forgotten. Currently, we are trying to recall the combination code for our bags, which we have forgotten. The generation of viable solutions may be "blind" or confined to a specific subregion of the state space (identified by significant dates), and it is crucial to acknowledge the likelihood of this occurrence. Moreover, we possess the capability to utilise a specific metric, termed a heuristic function, which signifies proximity to an acceptable solution.

Artificial intelligence (AI) widely employs the fundamental concept of viewing problem-solving as navigating a state space. This strategy enables robots to derive answers by adhering to clearly specified rules, providing a systematic and mathematical framework for describing complex issues. Numerous conventional artificial intelligence systems, including game-playing algorithms and robotic planners, rely on this paradigm, which allows them to make decisions and solve problems effectively by following predefined rules and structures.

A state is a notion that delineates a specific configuration of the world at a given moment. The state space comprises all theoretically achievable states based on the beginning conditions. The problem-solving process commences at an initial point and progresses toward a specified endpoint. Between these two states, a multitude of intermediate states exist. The states are interconnected through actions or operators that transform one state into another.

Issue resolution can be conceptualised as an exploration within a state space, a fundamental premise in both classical and symbolic artificial intelligence. An artificial intelligence system does not spontaneously generate a solution; instead, it methodically advances from an initial condition toward a desired goal by exploring the numerous possible options. In this image, intelligence is defined as the ability to search efficiently.

At any given moment, a state represents a certain configuration of the environment or the problem. The state space refers to a comprehensive collection of all possible states that may arise. An initial state serves as the system's starting position, then applying a set of permissible operations, commonly known as rules, actions, or operators, to achieve a desired state.

Consider the task of solving a puzzle like the 8-puzzle, which involves sliding tiles. Each arrangement of tiles represents a state. Relocating a tile generates a new state. To accomplish

the mission, it is essential to explore the path from the initial configuration to the desired configuration. The artificial intelligence must analyse alternative moves, evaluate them, and persist until achieving the goal.

Means-Ends Analysis

The Means–Ends Analysis, or MEA, is regarded as one of the most significant strategies in the annals of cognitive psychology and artificial intelligence. The contributions of Allen Newell and Herbert Simon to the General Problem Solver (GPS) in the 1950s and 1960s significantly influenced the advancement and dissemination of the Model Execution Algorithm (MEA). This model provides a systematic representation of the progression of intelligent agents from their current state to a targeted goal. Recognising the disparity between the present condition and the desired condition, followed by the selection of activities that mitigate this disparity, is essential for problem-solving. This philosophy is based on a simple yet profound concept. With persistent efforts to bridge the gap, the agent will ultimately get the intended outcome.

A robust correlation exists between the origins of means-ends analysis and the initial endeavours to understand human cognitive processes through computational frameworks. Newell and Simon posited that human reasoning can be defined as a sort of symbol manipulation. If this is the case, it should be possible to create computer programs that follow similar procedures, such as those that can solve complex problems by mimicking human reasoning processes. MEA emerged from comprehensive research as a pragmatic approach for directing search operations in intricate issue domains. Rather than indiscriminately exploring all conceivable choices, the system would utilise its knowledge of the distinctions to progress along a deliberate path.

An issue is frequently characterised by numerous components within the MEA framework. The components comprise an initial state, a terminal state, and a collection of operators or operations capable of transforming one state into another. The solver compares the current state with the objective and identifies the distinguishing qualities between them. These distinctions then influence the choice of an operator. Should the operator be applicable, the state will alter, and the system will progress toward the objective. If implementation is unfeasible due to the lack of specific requirements, those requirements will be regarded as sub-goals that must first be fulfilled.

Sixteen: A key element of MEA is the establishment of sub-goals. It is uncommon for a singular direct action to resolve intricate circumstances. The most effective way to achieve success is to establish a hierarchy of intermediate accomplishments. By temporarily diverting attention to these minor activities, the agent can decompose the problem into more manageable

components. Upon achieving the subgoals, the overarching objective will become more attainable.

Consider a simple illustration from everyday life. For instance, consider an individual wishing to go from one city to another. The positioning is the fundamental element that distinguishes the current state from the desired state. The most apparent choice may be to take a flight or a train. To achieve this, you must satisfy certain conditions, including acquiring a ticket and arriving at the station or airport. These needs lead to the establishment of new sub-goals, such as planning the travel itinerary and ensuring all necessary documents are in order. By sequentially accomplishing each task, the tourist will ultimately reach their objective. The MEA exemplifies this mode of pragmatic reasoning.

Means-ends analysis has proven to be exceptionally beneficial in artificial intelligence, as it significantly reduces the search space. In some instances, the quantity of possible answers can be exceedingly vast. Investigating every possible option would be either inefficient or unfeasible without some guidance, such as prioritising the most impactful strategies or focusing on specific areas of concern. The MEA directs its efforts only towards activities deemed relevant for mitigating inequities, such as targeted outreach programs and resource allocation strategies that address specific community needs. This exemplifies a heuristic search, which manifests in this manner.

The General Problem Solver provided a tangible illustration of implementing MEA. The design objective of the Global Positioning System (GPS) was to create a global system capable of managing diverse tasks, including mathematical proofs and logic difficulties. It symbolically contained descriptions of states and operators. It would assess the present state against the objective, identify a discrepancy, and thereafter employ an operator to eliminate the difference upon encountering a challenge. Should any obstacles arise, it would establish sub-goals and thereafter pursue them in a recursive fashion, allowing for a structured approach to problem-solving that adapts to the challenges encountered. GPS demonstrated the potential for mechanising general cognitive notions, but with certain limitations.

Moreover, MEA had substantial implications for the philosophy of mind to contemplate. It specifically endorsed the perspective that intelligence could be elucidated through formal procedures. If robots and humans use similar strategies to reduce differences, it is conceivable that cognition is inherently computational. This concept significantly contributed to the advancement of functionalism and reinforced confidence in the feasibility of artificial intelligence, as it suggests that cognitive processes can be replicated in machines, thereby bridging the gap between human and robotic intelligence.

The transparency of MEA is an additional benefit of its use. The reliance on explicit comparisons and rule application allows the system to consistently elucidate the rationale behind its actions. This competence is advantageous in domains where justification is crucial, such as medical diagnosis or planning. Symbolic techniques, such as MEA, differentiate themselves from specific modern machine learning models by rendering reasoning transparent, which is particularly important in fields like healthcare where understanding the rationale behind decisions can significantly impact patient outcomes.

To manage the generation of potential solutions, we can employ a method called Means-Ends Analysis (MEA), provided we have a function that evaluates the quality of the proposed solutions. When in a specific state, we will use a function to compute the difference or distance to a desired state, then apply a strategy to minimise this difference. For example, if our sample omelette is nearly as delectable as the perfect version, with the only distinction being insufficient sweetness, we should employ a "sweetening operator" to minimise the disparity between the two. We utilise the MEA technique iteratively, commencing from the initial state and progressing until we reach the desired state.

Problem reduction

The approach involves deconstructing a complex problem into a number of more manageable subproblems. For instance, if obtaining a driving licence is the challenge at hand, we should decompose it into two subproblems: successfully completing a theoretical examination and passing a practical driving test. It is very important to acknowledge that the simplification of numerous issues into a sequence of more manageable subproblems is necessary. This arises because unique state spaces must be defined for each subproblem, hence requiring the identification of various quality functions and distinct operators, which ultimately facilitates a clearer understanding and solution of the overall problem.

In the domain of symbolic artificial intelligence (AI), problem reduction is regarded as a fundamental methodology. Symbolic artificial intelligence (AI) posits that intelligence is the manipulation of explicit representations, including symbols, rules, and logical structures. In this setting, it is often difficult to directly achieve very complicated goals. Consequently, the system strives to break down a significant issue into several smaller, more manageable subproblems for incremental resolution.

In symbolic artificial intelligence, information is articulated in formal languages, including propositions, production rules, or logical relations. When a system is assigned a goal, it will seek activities capable of transforming the current state into the desired state. If the objective

is overly challenging, the programme will provide intermediate goals. Resolving these issues steadily progresses towards the final objective. The essence of problem -minimisation lies here. The General Problem Solver (GPS), developed by Allen Newell and Herbert Simon, serves as a conventional example. The GPS system employed a technique referred to as means–ends analysis. The system initially assessed the present state against the intended state, identified the disparity between the two, and ultimately selected an action to minimise the gap toward the desired state. If the action required certain criteria, the conditions established new subgoals. The first problem was divided into a hierarchy of more achievable tasks.

AND–OR graphs are a prevalent depiction of problem reduction in symbolic artificial intelligence.

Success at an AND node necessitates the resolution of all subproblems.

For an OR node, it is adequate to resolve any single alternative.

This representation enables the machine to structure search paths and identify the most efficient routes to attain a solution. This exemplifies logical reasoning, wherein specific conclusions necessitate many premises, yet others may logically derive from other potential possibilities.

This technique aligns seamlessly with symbolic artificial intelligence, as symbols may accurately communicate goals, sub-goals, and regulations. In numerous instances, the system can elucidate the rationale behind its conclusions, since it operates via a sequence of distinct thinking phases, which allows users to understand the decision-making process and the underlying logic that leads to specific outcomes. Symbolic methods possess several advantages over purely statistical approaches, one of which is their transparency, as they allow users to understand the reasoning process and the rules applied in reaching conclusions.

Nonetheless, problem reduction reveals numerous limitations of symbolic artificial intelligence. In real-world situations, it can be challenging to delineate distinct components due to their complexity, ambiguity, or interdependence. Selecting inappropriate subgoals may result in a waste of effort or lead to a dead end. The extent of the search space may expand significantly as the complexity of the problems escalates.

Despite these limitations, the notion of issue minimisation remains significant. Hierarchical goal decomposition remains common in modern planning systems, automated reasoning applications, and certain hybrid artificial intelligence models. A fundamental understanding of intelligence is that deconstructing difficult objectives into organised elements often makes them manageable.

Logic-Based Approach

John McCarthy, a pioneer of logic-based methodologies for artificial intelligence, contended that intelligent systems ought to be constructed on codified models of logical reasoning rather than as imitations of heuristic rules governing human cognition. McCarthy's findings appeared in the journal Artificial Intelligence. The subsequent disclosures rendered this concept innovative in the domain of computer science by the late 1950s. During that period, the imperative paradigm was the basis for the conventional methodology in the development and implementation of information systems. The technique involves delineating a program as a series of instructions that a computer must execute.

McCarthy asserts that a programmer developing an intelligent program should focus solely on the essential attributes of a solution, as these are paramount. In other words, he or she should specify the solution to the problem rather than concentrating on the method of obtaining it. A comprehensive issue resolver, usually referred to as generic problem-solving software, should be responsible for finding the solution. This method for the creation and deployment of information systems is feasible using the declarative paradigm as a basis.

To create software utilising the declarative paradigm that determines if two individuals are siblings, we can proceed as follows:

siblings (X, Y): − parent (Z, X) and parent (Z, Y),

It is comprehended as follows:

X and Y are siblings if there exists a Z who is a parent of both X and Y.

In our software, we specify only the necessary properties of a solution.

Logic programming and functional programming are primary methodologies within the declarative paradigm in artificial intelligence. Logic programming defines the necessary features of a solution as a set of theorems in a logical language, usually First-Order Logic (FOL). Numerous varieties of logical languages exist. A person adept at resolving universal issues uses these theorems to conduct reasoning consistent with the principles of inference in first-order logic. To resolve a specific issue with the solver, such as ascertaining whether John Smith and Mary Smith are siblings, we construct a hypothesis framed as siblings (John Smith, Mary Smith). The system endeavours to validate this hypothesis by using facts from its knowledge base and relevant theorems, including the previously mentioned definition of sibling relationships.

Lambda calculus, originating from the contributions of Alonzo Church and Stephen C. Kleene, underpins functional programming. The lambda calculus is a formal system of mathematical logic employed to define computational functions through highly structured expressions. In

functional programming, an advanced function delineates the essential characteristics of a solution. The elucidation of these terms is essential for a universal problem solver to assess the functions associated with them in accordance with the principles of lambda calculus.

In contrast to the first group, the second group's research efforts aim to establish models tailored to certain application domains. Consequently, the representations of domain knowledge are the foundation for these types of models. This subclass includes several techniques of knowledge representation, such as rule-based, structural, and mathematics-based approaches.

One of the most significant and historically influential traditions in artificial intelligence is the logic-based approach. The foundational principle of this perspective is that intelligent behaviour can be established by articulating knowledge in a formal logical language and applying inference rules to that knowledge. This perspective asserts that logic's particular methodologies can articulate thinking as a fundamental process of reasoning. For several decades, this concept has been a guiding principle in artificial intelligence research, impacting both practical systems and philosophical hypotheses regarding the nature of the mind.

An approach based on logic can be traced to mathematical logic, analytic philosophy, and early advancements in computing. Logicians have already established the capacity to automatically infer conclusions from premises by following stringent rules. With the advent of widely accessible digital computers, scholars began to contemplate the potential for these machines to autonomously do reasoning. Intelligence, defined as the capacity to derive appropriate conclusions from provided information, suggests that programming a computer with logical principles may allow it to function intelligently.

A critical part of this strategy is the representation of knowledge. A symbolic representation of the world is essential for the machine to process it. Logical assertions can describe facts, relationships, and general concepts. Subsequent to the establishment of the knowledge base, an inference engine will apply rules to derive new conclusions. The technique concludes that "Socrates is mortal" based on the premises "All humans are mortal" and "Socrates is a human." This illustrates how formal thinking can enhance the growth of knowledge.

Artificial intelligence extensively employs two fundamental logical systems: propositional logic and predicate (first-order) logic. Propositional logic pertains to unequivocal statements that can be classified as either true or false. Predicate logic surpasses other forms of logic due to its capacity to articulate characteristics, quantities, and connections among entities. The expressive capacity of predicate logic facilitated its selection as a basis for numerous artificial intelligence programmes, allowing these computers to handle complex informational frameworks.

Numerous examples illustrated the efficacy of the logic-based approach. It was a major contributor to automated theorem proving, the process by which machines endeavour to establish mathematical results. Furthermore, it was crucial in the advancement of expert systems, which encoded specialists' expertise into rule-based structures. During the planning and decision-making process, logical systems aided in identifying the actions necessary to transition from their existing state to the desired target. The rationale in each case adhered to methods that were both clear and verifiable, ensuring that stakeholders could understand and trust the decision-making process.

One of the most notable attributes of the logic-based method is its clarity and transparency. Since conclusions are derived by formal rules, it is generally feasible to elucidate the exact process by which a result was obtained. This method is attractive in disciplines such as engineering, medicine, and law, all of which prioritise justification significantly. The system's thinking can be examined, validated, and adjusted if necessary.

Furthermore, the logic-based approach reinforces the perspective that intelligence fundamentally aligns with rationality. This perspective depicts thinking as a systematic process governed by criteria of validity and coherence. This image aligns with philosophical traditions that prioritise reason as a distinguishing trait of the mind over emotion.

Rule-Based Knowledge Representation

After presenting their idea of cognitive simulation, Newell and Simon continued their exploration of cognitive process models throughout their research. The system comprises two essential components: production memory and working memory. From a psychological perspective, production memory corresponds to long-term memory, while working memory aligns with short-term memory. Long-term memory is a category of memory that retains knowledge through the application of processes and principles. A rule, in its fundamental essence, is articulated as follows: Upon fulfilling a specific condition, it is vital to execute a designated action. This action may either be a conclusion derived from the condition or a specific action executed within the system's environment.

Rule-based knowledge representation is regarded as one of the most important and often employed strategies in artificial intelligence (AI) for knowledge storage and manipulation. This method entails the representation of knowledge through rules, typically formulated as IF–THEN statements. These rules enable artificial intelligence systems to resolve issues, draw conclusions, and render judgements akin to human reasoning.

The condition, referred to as the antecedent, and the action or conclusion, known as the consequent, are often components of a rule. Upon meeting the condition, the system will execute the action. For instance, if a patient exhibits coughing and elevated temperature, it is plausible that they are experiencing an infection. These principles can be integrated to establish a knowledge base, a repository of information pertaining to a certain domain, such as medical symptoms and their potential causes, which can aid healthcare professionals in diagnosing conditions more effectively.

Rule-based representation became increasingly popular, particularly in the domain of expert systems, where the aim was to encapsulate the experience of professionals. In the realm of medical diagnostics, systems such as MYCIN employed numerous factors to formulate therapeutic recommendations. Using these guidelines, the system successfully emulated expert-level decision-making.

The inference engine is a crucial element of rule-based systems. The inference engine employs a series of logical procedures to determine which rules should be activated. The two principal ways of reasoning are as follows:

1. **Forward chaining: it starts with** known facts and applies rules to generate new conclusions until the goal is reached.
2. **Backward chaining** is a reasoning method that starts with a goal and works backward to see if available facts support it.

This framework enhances the transparency of rule-based systems. The rescinded regulations may be traced, and the decision-making process can be elucidated. Consequently, rule-based artificial intelligence is sometimes considered more interpretable than other modern machine learning models.

Rule-based knowledge representation exhibits several beneficial characteristics. It possesses a modular architecture, is easily understandable, and closely resembles human reasoning. Implementing new regulations allows for the integration of more information without necessitating a comprehensive modification of the system, which enhances its adaptability and efficiency in responding to evolving knowledge in various domains. In domains characterised by a definitive and coherent body of knowledge, they operate effectively.

Structural Knowledge Representation

Gilbert Ryle's thesis posits that our classification of knowledge comprises declarative knowledge, which pertains to static facts ("knowing that"), and procedural knowledge, which relates to the execution of tasks ("knowing how"). Both categories of information are essential

for our comprehension of the world. A genealogical tree exemplifies declarative knowledge, while a heuristic algorithm, which emulates human problem-solving, represents procedural knowledge. Descriptive knowledge is characterised by the use of structural models for knowledge representation. They frequently assume the form of hierarchical systems akin to graphs. Initially, it was believed that they could be employed for Natural Language Processing (NLP), but it was subsequently revealed that they could also be applied in other domains of artificial intelligence, such as computer vision, robotics, and recommendation systems.

This method uses frameworks like conceptual graphs, frames, semantic networks, and scripts to articulate knowledge. Such structures exemplify the relationships among events, properties, and objects. The objective of structural representation is to create a map of interrelations within a domain, rather than simply transmitting "facts".

The semantic network is among the first of its kind and is frequently utilised. In a semantic network, concepts are symbolically represented as nodes, while the connections between them are illustrated as links. For instance, one can assert, "A dog is a creature" or "A dog has four limbs." The transmission of attributes is feasible via this network. Like other animals, dogs also breathe. This hierarchical structure promotes effective reasoning and prevents redundancy.

Marvin Minsky was the first to introduce the frame, which can be conceptualised as a data structure that facilitates the expression of a concept. It comprises various components known as slots, each of which retains a specific piece of information or value. An example of this would be a vehicle frame that includes slots for attributes such as colour, engine type, manufacturer, owner, and speed. This system can either populate or modify these slots when new information is accessible. Default settings for frames are also given. In the absence of specific indications, the system will assume that a standard automobile possesses four wheels. Consequently, artificial intelligence may formulate swift, logic-driven conclusions without the necessity of explicitly revealing every detail.

A frame operates comparably to a data structure in representing prototypical settings. It contains properties and corresponding values. A "vehicle" framework may include fields for the owner's name, engine kind, colour, and speed. Frames can be updated when new information emerges, and they permit the use of default values. Consequently, they are effective in transmitting widely accepted knowledge.

It is customary to employ scripts to elucidate standard sequences of events. Using a restaurant as an example, the procedure may include the following steps: entry, order placement, dining, payment, and exit. Artificial intelligence systems can more effectively understand expectations

and forecast potential future events due to scripts, which outline typical customer behaviours and preferences in a restaurant setting, such as common menu choices and peak dining times.
Structural knowledge representation has numerous advantages. Besides enabling inheritance and improving reasoning efficacy, it provides a clear framework of information. Furthermore, it closely resembles the manner in which people conceptually structure knowledge, which is advantageous for both natural language interpretation and expert systems.

Mathematical Linguistics Approach

Artificial intelligence encounters several fundamental obstacles, one being the development of a description of the physical world that may be employed for the reasoning of intelligent systems. At times, in the realm of artificial intelligence, our focus is more on constructing systems for sentence interpretation than on developing the sentences themselves. Fortunately, formal automata can be employed to achieve this objective. In mathematical linguistics, they are delineated for the purpose of generating grammars. A formal automaton can be characterised by evaluating whether an utterance is a sentence in a specific language. This represents the minimum scenario. In other words, does the phrase possess a definition that aligns with the grammatical principles pertinent to the language in question? A pair of generative grammars and formal automatons can indeed be considered a fundamental physical symbol system.

Mathematical linguistics fundamentally examines the application of formal models to delineate grammar and semantics. The analysis of sentences is conducted based on specified criteria rather than relying on intuition or informal interpretation. This enables computers to understand, produce, and modify language systematically.

A notable addition to this methodology is formal grammar theory, especially the research of Noam Chomsky. This constitutes a significant contribution. He proposed that languages can be defined by rule systems, such as phrase structure grammars, and classified into many types, including regular, context-free, context-sensitive, and others. Using these grammars, artificial intelligence systems can ascertain the structure of a sentence and validate its correctness. Phrase structure grammars are formal systems designed to delineate the construction of sentences within a language. They play an essential role in artificial intelligence, specifically in natural language processing (NLP), parsing, and computational linguistics. Structural grammars are a category of grammar that delineates principles governing the combination of smaller units, such as words, to create larger units, such as phrases and sentences.

The essential idea is that a hierarchical structure exists within the process of language. Noun phrases and verb phrases exemplify the constituents that comprise a sentence. A sentence is not merely a collection of words placed together. Structured grammars endeavour to represent this configuration through the application of rewrite rules.

The use of logic to convey meaning is another crucial element. Utilising predicate logic, or first-order logic, enables the transformation of assertions into mathematical expressions. It is feasible, for example, to symbolically represent the statement "All humans are mortal", thus allowing machines to perform deductions and derive conclusions. Artificial intelligence can deduce facts articulated in language using this logical representation.

Additional elements of the mathematical linguistics approach encompass automata theory and formal languages, both of which facilitate the modelling of symbolic string processing. To understand grammar and identify language, computational models such as finite-state machines (which process input strings in a linear fashion), pushdown automata (which can use a stack to manage additional information), and Turing machines (which are theoretical devices that manipulate symbols on a tape) are necessary. Automata theory is directly linked to the study of formal languages and grammars. A language is merely a compilation of strings derived from symbols. Automata are effective instruments for ascertaining the association of a certain string with a specific language. Numerous applications of artificial intelligence pertain to the interplay between machines and languages.

The accuracy of this method is one of its most notable features. Explicitly established rules can reduce ambiguity and enhance transparency in thinking. It offers functionalities including parsing, machine translation, question answering, and dialogue systems.

Neural Networks

Cognitive scientists denote brain networks as connectionism. The objective of neural networks was to develop intelligent systems as networks of nodes, with each node serving as a simplified representation of a neurone. The distinction was between a computer analogy and a brain analogy, contrasting the implementation of artificial intelligence systems as conventional computer programs with their emulation of neural systems. In 1951, Marvin Minsky and Dean Edmonds constructed the inaugural artificial neural network, illustrating a rat traversing a maze. The network was referred to as the SNARC (Stochastic Neural Analogue Reinforcement Calculator). This investigation constituted the basis for Minsky's dissertation at Princeton in 1954. Consequently, one developer and key figure in symbolic artificial intelligence initially

focused on neural networks, establishing the groundwork for its computing implementation, which later influenced the development of more advanced AI systems and algorithms.

Nodes, referred to as "units", constitute the fundamental components of a neural network. These nodes are comparable to neurones located in the brain. Each node is tasked with generating an output signal while concurrently receiving several input signals. The output of one node might serve as the input for another node due to their interconnected configuration. It is customary to use real numbers to denote input and output values. Each link is associated with weights, which are real values. The weight of a link intuitively signifies the extent to which one node influences the output of another node. It is standard procedure to calculate the weighted total of the input values, yielding an output of either 1 or 0, contingent upon whether the sum surpasses a specified threshold. The output of each node is a linear function of its inputs.

There are two principal methodologies for artificial intelligence, corresponding to the models of the human brain and evolutionary processes. The artificial intelligence model that emulates the brain aligns with symbolic AI and is the foremost subject in the domain of artificial intelligence. This approach is considered the macroscopic perspective of artificial intelligence because it involves significant mathematical and algebraic abstraction. The foundational phases of the classical artificial intelligence (AI) model occurred during the 1950s and 1960s. This model was originally designated as the classical model. Initially, it was referred to as symbolic AI or rule-based intelligent systems. The fundamental premise of the notion was the assumption that human intellect could be replicated through logical reasoning and symbol manipulation. This understanding laid the foundation for the concept. Initially, it was believed that artificial intelligence systems could resolve problems by the application of formal logic, predefined rules, and symbolic representations. The Logic Theorist and the General Problem Solver exemplify information processing systems under this category. This model was essential in establishing the groundwork for artificial intelligence research; yet, it was hindered by several substantial flaws and limits that obstructed its advancement and precluded its practical application, such as its inability to handle uncertainty and its reliance on rigid rule-based systems. Although it was crucial, it could not be implemented in practice. Several challenges and limitations hindered its application in the actual world.

In numerous instances, it is based on the Physical Symbol System Hypothesis (PSSH). The Classical or Symbolic model of Artificial Intelligence (AI) presents a description of cognition that is intriguing yet potentially contentious from a philosophical perspective. This concept posits that intelligence primarily involves the manipulation of physical symbols according to explicit formal rules, akin to how a computer program executes an algorithm. The purpose of this model is delineated. This method's failure to consider intentionality—characterised by "aboutness" and genuine comprehension inherent to the human mind—led to early achievements in areas like logic and game-playing. Nonetheless, it precipitated a substantial conceptual crisis. An intriguing illustration of this issue is the argument given by philosopher John Searle in the Chinese Room scenario. It highlights a significant gap between simple syntactical manipulation and profound semantic understanding, demonstrating that while a system may process language, it lacks true comprehension of meaning and context. The PSSH posits that the mind functions as hardware while cognition operates as software, forming the basis of conventional artificial intelligence. In this system, discrete and abstract symbols represent concepts, and reasoning is performed by following a programme (rules) that dictates how these symbols are combined and modified. The system, due to its syntactic nature, focuses solely on the formal attributes of the symbols, encompassing their shape, structure, and positioning. It inherently possesses a syntactic characteristic. For example, a system may accurately process the statements "Socrates is a man" and "All men are mortal" to derive the conclusion "Socrates is mortal." The system exhibits signs of intelligent behaviour when evaluated from a functional perspective, as it can draw logical conclusions based on the information provided, even if it does not truly understand the concepts of "man" or "mortal." Enquiring if the system truly understands the concepts of "man" or "mortal" presents a philosophical challenge. In classical artificial intelligence, the objective is to understand the structure, commonly referred to as syntax. However, it seems to be completely oblivious to the substance or meaning, which is referred to as the semantics of the language.

The classical model has a substantial flaw: it fails to integrate arguments that should be considered common sense. This constitutes a fundamental defect. Although these systems excelled in controlled and formal situations, they faltered in real-world scenarios when knowledge was ambiguous or inadequate, leading to failures in decision-making and problem-solving that highlight their limitations compared to human cognition. We will never replicate the human ability for adaptable and context-dependent understanding via symbolic systems. An artificial intelligence based on rules would struggle to differentiate between literal and

metaphorical meanings and to manage linguistic ambiguity. This would provide a difficulty for the AI.

Concerning the representation of knowledge, another dilemma persists. Utilising the classical approach required the laborious encoding of all data into symbols and rules prior to its use. Consequently, the systems were unable to acquire knowledge from their experiences, rendering them inflexible. In many instances, individuals failed when presented with facts that were either unfamiliar or unforeseen. This is due to the previously described qualities. In a context characterised by constant change, the frame problem refers to the challenge of identifying which facts are pertinent to the issue at hand. The term "framing challenge" is occasionally employed to denote this particular obstacle. Moreover, conventional artificial intelligence has challenges with the limitations of its computational capacities. Early computers, constrained by limited memory and computing capacity, struggled to exhibit intelligence through symbolic representation. This was due to their deficiency in both traits. In complicated, real-world situations involving perception, ambiguity, or sensory data—domains where humans inherently excel—symbolic reasoning is recognised as a sluggish reasoning method. This is due to the complexity of these events. The inability to acquire new information constituted a significant deficiency. The conventional iteration of artificial intelligence was entirely reliant on human programmers to furnish it with the required rules and knowledge it depended upon. Unlike contemporary machine learning systems, which can evaluate data and autonomously evolve, these systems lack the capacity to adapt to novel situations or enhance their performance via experience, resulting in a rigid approach that limits their effectiveness in dynamic environments. In conclusion, it is essential to acknowledge that the conventional method failed to account for the psychological and biological factors influencing an individual's intelligence level. Historically, individuals perceived thinking as exclusively verbal and intellectual, neglecting the roles of emotions, intuition, and perception. This resulted in a limited and mechanistic conception of intelligence, inadequate for encompassing the complexities inherent in human cognition, which include emotional intelligence, creative problem-solving, and the ability to navigate social interactions.

The capacity of artificial intelligence to emulate human reasoning has been a subject of inquiry for philosophers and society. A critical inquiry pertains to the capacity of robots for cognition and the assessment of their proficiency in executing designated tasks. Alan Turing proposed a test to assess the cognitive capabilities of a computer to examine the question of artificial

intelligence's competence. He claimed that if the machine meets these criteria, we may regard the AI as equivalent to a human being.

Demonstrate what you obtained

1 Describe the concept classical model of AI.

It was during the 1950s and 1960s that the classical model of artificial intelligence (AI) was developed. This model, initially referred to as symbolic AI or rule-based AI, is also known as the classical model. The idea was based on the assumption that it was possible to simulate human intelligence by employing logical reasoning and manipulating symbols.

2 Describe weakness classical model of AI.

There is a serious weakness in the classical model, and that flaw is that it does not integrate any reasoning that should be considered common sense. Although these systems were able to perform admirably in controlled and formal settings, they were not successful in situations that occurred in the real world, when knowledge was either ambiguous or insufficient.

3 Elucidate the significance of the Alan Turing Test.

In the modern era of artificial intelligence, the Turing Test remains relevant. It functions as a primary standard for evaluating conversational AI systems, such as chatbots and virtual assistants. Programs like ELIZA (1966), a direct descendant of Turing's vision, seek to emulate authentic human discourse. The test has also influenced research in Natural Language Processing (NLP), machine learning, and human-computer interaction, inspiring systems that can comprehend, reason, and respond efficiently.

4 Describe significance John Searle

John Searle (born 1932) is an American philosopher renowned for his contributions to the philosophy of mind, language, and consciousness. He is most renowned for his critique of Strong Artificial Intelligence (AI) via his renowned Chinese Room Argument (1980).

Knock your Brain

1during the -----and 1960s that the classical model of artificial intelligence (AI) was developed.

(1950s,1920,1930,1910)

2 symbolic AI or rule-based AI, was also known as the---------.

(classical model, postmodern, modern universal)

3 PSSH stand for

(Physical Symbol System Hypothesis, Physical System Symbol Hypothesis, Philosophical System Symbol Hypothesis, Physiological System Symbol Hypothesis)

4 ---------- proposed a test to evaluate a machine's reasoning power, asserting that if it meets these criteria, we may regard the AI as equivalent to a human.

(Alan Turing, John Searle, Plato, Locke)

5----------is a subfield of ethics that focusses on the practical application of ethical concepts

(Truth, Correspondence, Justification Applied ethics)

Unleash your potential

1. Analyze the notion Chinese Room Experiment
2. evaluates the concept Turing Test